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Smart Agriculture and Next-Gen Sustainability: Harnessing Big Data and Machine Learning for Carbon Sequestration Prediction with Blockchain-Powered Carbon Credit Trading

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Abstract

As climate change accelerates, there is an urgent need for solutions that balance ecological responsibility with economic incentives. While capping carbon emissions is widely recognized as essential, it remains a challenging task to quantify carbon sequestration correctly and ensure complete transparency in carbon credit markets. The increasing demand for effective carbon sequestration and transparent carbon credit trading demands an innovative approach using these advanced technologies. The present research focuses on the application of big data using Kafka for parallel data streaming in a distributed environment, together with ML models to optimize the prediction of carbon capture, while blockchain technology provides security and transparency in transactions involving the carbon credit market. These findings indicate that this interdisciplinary framework will indeed improve the accuracy and scalability of carbon sequestration predictions, building trust and accountability in carbon trading to support a more sustainable and economically viable future.

Keywords: Big Data, Machine Learning, Blockchain Technology, Kafka, Distributed Computing, Smart Agriculture, Environmental Sustainability, Carbon Credit Trading

1 Introduction

The global climate crisis poses an unprecedented challenge, threatening ecosystems, economies, and societies worldwide[1]. Human activities in the past century, such as industrialization, deforestation, and the extensive burning of fossil fuels, have dramatically increased the concentration of greenhouse gases (GHGs) in the atmosphere, particularly carbon dioxide (CO₂). These growing levels of greenhouse gases are the main driver of global warming, associated with increased global temperatures, more frequent and intense weather events, and widespread environmental degradation. Changes in climate manifest themselves in the form of melting polar ice caps, sea level rise, prolonged droughts, and intensification of natural disasters such as hurricanes and wildfires. These disruptions, both environmental and social, include loss of biodiversity, reduced agricultural yields, increased food insecurity, water scarcity, displacement of communities, and negative health outcomes.

In the face of the urgency of the climate crisis, the international community has mobilized through frameworks such as the Paris Agreement[2], which aims to limit the global temperature rise to below 2 °C above preindustrial levels. Achieving this goal requires a two-fold approach: (1) reducing ongoing GHG emissions through transitions to renewable energy, energy efficiency, and sustainable practices, and (2) actively removing CO₂ from the atmosphere. The latter is where carbon sequestration plays a vital role. Carbon sequestration involves the capture and long-term storage of atmospheric CO₂ in natural sinks such as forests, oceans, and soils. Among these methods, soil carbon sequestration has become particularly effective and scalable. This includes storing CO₂ in soils to mitigate climate change, improve soil health, enhance agricultural productivity, and promote regenerative land use practices. However, complete realization of the full potential of soil carbon sequestration requires appropriate measurement and verification mechanisms, which are currently limited by technical and methodological challenges.

1.1 The Impact of Carbon Sequestration on Reducing Climate Change

Soil carbon sequestration is a process where plants absorb atmospheric CO₂ through photosynthesis, which is then stored in the soil as organic carbon[3, 4]. This natural process helps offset GHG emissions, contributing to the reduction of atmospheric CO₂ levels. In addition to its role in mitigating climate change, soil carbon sequestration has multiple cobenefits, such as improving soil structure, enhancing nutrient cycling, increasing water retention, and fostering biodiversity. This makes soil carbon sequestration, besides being one of the many solutions to the climate problem, another key component in sustainable agriculture and land management practices.

It is about regenerative agriculture, a suite of farming principles designed to regenerate soil health, that soil carbon sequestration is most valued.

Practices like crop rotation, cover cropping, no tillage, and agroforestry enhance not only the sequestration of more carbon into the soils but also improve soil fertility and build resilience against weather extremes. These practices therefore improve the soil organic carbon and in turn make the ecosystem more productive and sustainable

to resist the impacts of climate change. However, one of the key challenges to leveraging soil carbon sequestration on a global scale is the ability to accurately measure how much carbon is being captured and stored. The amount of carbon sequestered can vary significantly based on several factors, including soil type, land management practices, vegetation cover, and local climate conditions[5]. Traditional methods of measurement involve soil sampling and laboratory analysis; most of these operations are very labor-intensive and time-consuming, often not with the required precision for large-scale application. As a matter of fact, all these methods have obvious limitations in geographic scope, which raises difficulty for their application at landscape or regional level.

1.2 Challenges in Measuring and Predicting Soil Carbon Sequestration

Prediction of soil carbon sequestration is a very complex task, given the number of variables that interact to influence it. Soil properties, climate conditions, vegetation type, agricultural techniques, and previous land use patterns will continue to become critical determinants in the amount of carbon that can be sequestered by any given ecosystem. Current predictive models are usually incapable of considering the full range of such variables, introducing inaccuracies into the estimates for carbon sequestration. Besides, soils are also highly variable in space—carbon content can be very different within one meter of area—further complicating the measurements.

Another major challenge is the lack of standardized methods for verifying carbon sequestration at scale. Verification is critical for the carbon credit market, where organizations purchase carbon credits to offset their emissions[6, 7]. Carbon credit is a market mechanism that fosters reductions by crediting a firm or an individual for capturing or reducing any form of emissions. In reality, the value of the credit calls for proper and transparent measurements of carbon sequestration. If carbon credits are not verified, they cannot be scaled up, and their integrity will be undermined, negating scaling up carbon sequestration initiatives or global emissions reduction.

1.3 Approach in solving this challenge

One of such projects that meets these challenges head-on is titled "Big Data and ML-Driven Carbon Sequestration with Blockchain-Enabled Carbon Credit Trading." The following solution will make use of three state-of-the-art technologies—big data, machine learning, and blockchain—while working their way around current methodologies' setbacks in prediction and verification of soil carbon sequestration.

Big Data for Real-Time Environmental Data Processing: It utilized Apache Kafka, a distributed streaming platform ingesting and processing huge volumes of real-time environmental data [8]. It provides for swift and straightforward data inflow from all sorts of sources, such as automated weather stations, remote sensors, satellite imagery, and many others, to assure that the latest relevant information is available at any moment in time. This real-time data stream is extremely useful in making correct predictions of carbon sequestration, as environmental variables like temperature, precipitation, and soil moisture levels are always changing.

Machine Learning for Improved Forecasts of Carbon Sequestration: The project will apply machine learning algorithms in the quest for improved forecasts of carbon sequestration.

ML models draw patterns and correlations among a range of variables by leveraging big environmental datasets captured via Kafka. In return, this brings more realistic predictions of carbon sequestration potential across alternative ecosystems. Its important applications will also result in the optimization of land management practices for diverse stakeholders through enhancement in their carbon capture and promotion of sustainable agricultural practices. Blockchain for transparent carbon credit trading involves the use of blockchain technology to ensure credibility and transparency in the carbon credit market. Blockchain's decentralized and immutable ledger provides a secure and transparent platform for recording carbon credit transactions [9]. Each transaction is verified and irrevocably recorded, thus preventing fraud and ensuring all parties have a trustworthy view of the data. This has been one of the essential sets of transparency that ensures the carbon credit market sets the record straight and encourages organizations to effectively capture carbon.

The combination of these technologies in this study therefore creates a robust framework that can boost accuracy and increase much-needed transparency in carbon sequestration predictions and Carbon Credit trading. Indeed, an integrated system would raise the accuracy of environmental data; it would engender trust and accountability in the carbon credit market and involve the largest number of people in carbon sequestration projects. This could be a work of immense potential regarding impact on climate action and economic practices. Using improved predictions of soil carbon sequestration, the stakeholders can offer optimizations w.r.t land management practices which would result in effective carbon capture. The blockchain-based carbon credit trading platform enforces this; hence, this makes carbon credits credible to therefore increase participation in carbon offset programs, hence making a better carbon market. "This approach links economic incentives with environmental responsibility, encouraging sustainable practice while contributing to global emission reductions.

2 Related Works

Saraji et al.[10] presented a holistic Blockchain-based Carbon Credit Ecosystem designed to enhance the efficiency and transparency of carbon markets by bringing forth greater liquidity, accessibility, and standardization. The proposed system incorporates all key stakeholders, such as carbon credit "Generators" like renewable energy projects and carbon sequestration initiatives, and "Consumers" like industries that emit carbon. Accredited "Validators" play a critical role ensuring carbon credits are properly validated before tokenized on the Blockchain. These credits are converted into digital tokens that can be traded on a decentralized exchange where token prices are determined by market forces, such as supply and demand. The ecosystem also has a "buy and burn" model that allows participants to retire their carbon tokens, with every successful burn resulting in the issuance of a non-fungible token (NFT) as a certificate of carbon offset. The four key smart contracts powering the ecosystem include the registry, which records the stakeholder information; a minting contract that issues,

manages, and creates carbon tokens; a multi-signature contract responsible for validator approvals; and the automated market maker (AMM) contract, responsible for trading carbon tokens and providing rewards to liquidity providers. All these add up to form a fair, decentralized, and transparent way of trading and retiring carbon credits, thus increasing the total effectiveness of carbon markets in tackling climatic change.

Li, Tong, et al.[11] explored the integration of remote sensing, machine learning, and mid-infrared spectroscopy (MIR) as an innovative approach to accurately measure soil organic carbon (SOC) concentration and estimate SOC stock. Their work highlights how this combination of technologies offers cost-effective, scalable, and precise solutions, addressing the growing need for accurate SOC measurement to promote sustainable agricultural practices. They report initial success in Australia, where optimized precision sampling techniques and the use of MIR have significantly reduced SOC measurement costs for farmers. This integration has also allowed for more frequent monitoring of SOC, enhancing the ability of farmers to track soil health over time. Although the authors present preliminary findings based on 36 sample datasets, they acknowledge the need for more extensive data to develop depth-specific models and support per-farm analyses. Moreover, the authors highlight the commercial potential of SOC measurement technologies, which could lower entry barriers for farmers into carbon markets and create economic opportunities. Early collaborations with carbon developers and corporations indicate a strong interest in adopting these technologies within Australia's emissions trading system. The study underscores the potential of these advanced methods to stimulate regional economies, provide valuable data for national soils databases, and foster collaboration between various stakeholders to drive sustainable farming practices and contribute to global climate change mitigation efforts.

Van Der Voort et al.[12] developed the Wageningen Soil Carbon STock pRotocol, an extensive method designed to estimate SOC stock across various agricultural soil types at reduced costs and time. The main steps of this protocol include the selection of spatial covariates from satellite data, the identification of sampling locations, SOC measurement using field-based soil spectroscopy, training a model on the collected data, and estimation of carbon stock with uncertainty estimates. Testing of the Soil-CASTOR protocol was done on farms in Arkansas and Iowa that comprised an area of about 235 hectares of silty loam and silty clay loam soils. The authors provide a robust and efficient method to estimate carbon stock by integrating satellite data with the NIR scanner measurements of SOC to determine the carbon stock up to an accuracy of 10 m. They estimated the carbon stock at the top 30 cm of soil in these fields to vary between 19 and 55 tons of carbon per hectare. This protocol thus applies widely to any carbon farming scheme and gives cheap and scalable access to monitoring any changes in SOC, thereby offering a route for sustainable land management practices.

Kyriakarakos G, Petropoulos T, et al.[13] discussed growing carbon farming importance and identified a need for scalable tools and tradable carbon credit certificates for the sustainability goals. Their study underlines the urgency of integration of several technologies and methodologies for the development of a low cost solution for small and medium sized farmers. The authors identify as a number of major important

technologies to be included. Among these are: Satellite and drone multispectral photography, Eddy Covacity, Electro-conductivity sensors contact and non contact and portable spectrometers for visible and near infrared spectral analyses. They further bring to the fore that the current level of technological advancements have made low-priced spectral sensors cost about EUR 25. Farm management information systems or FMIS, which integrate the farming and meteorological data, are also critical towards supporting carbon farming. Joining these technologies and the analysis with AI and Big Data analytics will give a novel and holistic approach for carbon farming. The authors tell the case that this set of tools and technologies is going to propel future carbon farming and catalyze scaling-up these practices across the globe.

Baklaga, Luka discussed the integration of AI with blockchain technology in the development of decentralized carbon markets, coming up with a new solution to curb carbon emissions[14]. Their research focuses on how AI plays a very key role in making the trade of carbon credits both reasonable and efficient, with blockchain ensuring transparency and security within transactions. The authors declare that the integration of AI models on predictive capabilities into a blockchain decentralized system has revamped carbon credit acquisition and delivery; hence, there would be more secure and traceable transactions. Another mechanism for enhancing the speed and efficiency of carbon credit trading is by using smart contracts on blockchain as it cuts the risk involved in fraudulent transactions. Of this approach, the researcher concludes there is qualitative resemblance in patterns between the spreads of neural networks and the blockchain technology, which has implications for the coming revolution for carbon markets. However, the study identifies some limitations: specifically, how to incentivize developers within a decentralized system and how to train accurate AI models when historical data are lacking. Some future research directions would include investigating regulatory frameworks and governance structures and how these could potentially include quantum computing in the development of decentralized carbon markets. It asks the world to move from issues of scalability, regulation, and implementation into the real-world application environment while demanding research and investment in AI and blockchain to have a sustainable carbon trading ecosystem.

Jawalkar et al.[15] propose a decentralized carbon credit trading system by applying blockchain and AI to accelerate the issuance of carbon credits with their verification and transfer process. This paper describes the deployment of a smart contract on the Goerli Ethereum test network using the Thirdweb platform, which is integrated with React.js for a user-friendly interface and MetaMask wallets for secure blockchain interactions. The system allows users to initiate transactions seamlessly while guaranteeing data integrity through auditing. They have some form of governmental organizations, an essential reason being that the EU Emission Trading System (EU ETS) sets targets over emission reduction and regulates a carbon market, but again, it is project verification-a Clean Development Mechanism-CDM or Gold Standard-that verifies emission reduction. The authors explain how AI and blockchain system together increase the transparency of the system, promote security and efficiency in trading carbon credits, thus improving upon more sustainable practices for mitigating emissions.

Acharya, Umesh, et al.[16] proposed a holistic approach for measuring soil carbon (C) by integrating laboratory analysis, in situ methods, and remote sensing techniques to overcome accuracy, cost, and scalability challenges. They reason that the recent approaches do not make the best use of the existing techniques as lab approaches are more precise, expensive, and results only come at one point in space while the in-situ approaches cover much space, but the accuracy is very poor. The authors then consider formulating a portable sensor-activated tool that measures all such significant properties like soil color, density, moisture, strength, and near infrared reflectance of soil at efficient data gathering with automatic saving in cloud platforms. These properties are leveraged by the machine learning algorithms in the device to predict soil C content over an entire field. Besides, they highlight the requirement of drone and satellite imagery to map soil properties over large areas while addressing the accuracy limits that exist today by pointing out critical soil characteristics influencing carbon content. Maybe, this new framework will allow data integration from handheld devices, drone imagery, and satellite observations into the detailing of soil C maps at the field, farm, or regional level. This is a methodology that enhances the measurement of soil C, promotes continuous monitoring of, and assessment of soils' health, and is promising in future research in data-driven agriculture.

Kalaiselvan, S. A., et al.[17] have proposed a new paradigm in carbon credit trading by developing a Blockchain-based Carbon Credit Marketplace to address the inefficiencies of the existing systems. This system makes use of a decentralized ledger, which ensures transparency, immutability, and traceability in carbon credit transactions, thereby increasing trust among participants. The mechanism of smart contracts makes possible the execution of transactions in an automated and very cost-effective manner without any intermediate parties. According to the authors, this is a decentralized governance model in which active stakeholders include verifiers of the project, the industry of energy, NGOs, and governments, who work with the citizen to raise cooperation in decision-making. To that extent, the marketplace determines protocol standards on carbon credit creation, validation, and trade exchange to prevent fragmentation, characteristic of many traditional markets. These innovative features allow the authors to create a trustworthy and secure marketplace, eliminating problems such as over-crediting and facilitating their cause in global efforts through incentives for sustainable practices. Overall, it takes a step toward bringing positive change in the environment along the pathway toward a more efficient, transparent, and sustainable future carbon credit trading.

3 Proposed Work

Three significant elements are integrated into the proposed system with the aim of improving carbon sequestration and to enable carbon credit trading. First, we use Kafka in managing large datasets by parallel data streaming, which supports the processing of large extensive datasets in a distributed way while achieving real-time and scalable data handling. The second part makes use of the strong data that is computed by Kafka for use in ML algorithms for making further accuracy and depth predictions in carbon sequestration. Third, to make a trustful setup, it constructs a

robust blockchain. Carbon credits provide a kind of verification for all the operations in a fully secured environment, making this work successful towards greater predictability and boosting market transparency for an ever-cleaner green. These steps are illustrated in Figure 1, showcasing the integration of **Kafka for real-time data streaming**, machine learning for enhanced prediction accuracy, and blockchain for secure and transparent carbon credit operations.

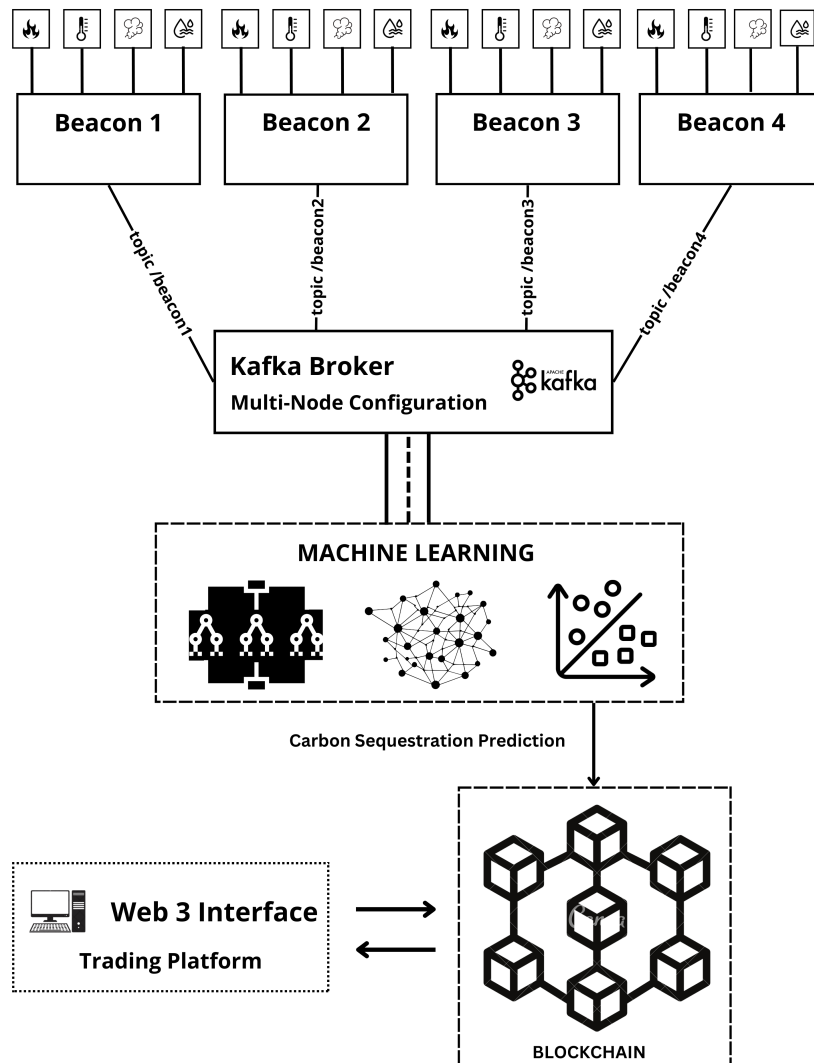


Fig. 1 Architecture Diagram

3.1 Big Data Pipeline: Leveraging Kafka for Real-Time Data Streaming and ML Model Integration

In the areas of environmental informatics and eco-informatics, the creation of an all-encompassing Big Data pipeline that will use Apache Kafka for a real-time data stream greatly benefits in terms of scale, fault tolerance, and seamless integration with machine learning (ML) models according to references cited in [18, 19]. Running as a distributed commit log, Kafka allows for great and ongoing data transfer and communication between disparate sources. It organizes and allows independent processing of geographically distributed data through the generation of output into Kafka topics based on ecological regions or designated environmental parameters. Figure 2 demonstrates a pictorial representation of how this system works. This methodology promotes significant concurrency, allowing multiple machine learning models to concurrently access data from different topics, thereby supporting parallel computation and delivering real-time insights into environmental phenomena.

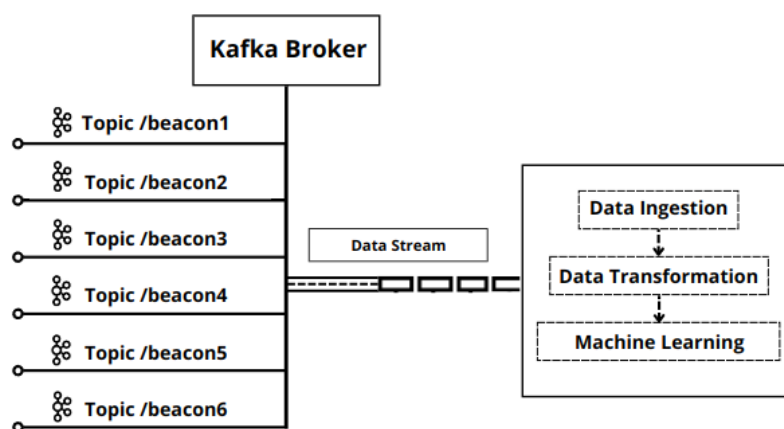


Fig. 2 Data ingestion from Kafka to Machine Learning Models

Kafka's role in the pipeline of eco-informatics is quite important because it can process tremendous amounts of data with negligible latency. The partitioned architecture inherent to Kafka makes it possible to manage large volumes of data by dividing the data into partitions, allowing multiple consumers to process the data at the same time. This does not only make the system more efficient but also ensures that the system scales smoothly as the volume and complexity of environmental data grow. The parallel reading capabilities of Kafka allow machine learning models to read from multiple partitions or topics in parallel, which may speed up the processing. This makes Kafka especially ideal for high-throughput applications such as real-time assessments of air quality sensor networks, predictive analytics on carbon sequestration across different ecosystems, real-time surveillance of deforestation trends by satellite images, and continuous assessments of oceanic temperature and acidity levels.

Another important feature that Kafka grants to the pipeline of environmental monitoring is fault tolerance. In its distributed nature, the architecture of Kafka is meant to tackle failures at different tiers. Data is duplicated at multiple brokers, and in case one broker fails, then the replicated data can be utilized at an alternative broker. In this way, both data reliability and consistency are assured, and it makes the system highly resilient-an important feature in handling critical environmental data. Inherent fault-tolerant mechanisms in Kafka include replica election for partition replicas of leader, acknowledgment-based write-operations for producers, as well as consumer offset managing against data loss during transportation and processing. This application would be critical for so many real-time applications working within data integrity requirements, such as a system that monitors climate changes or biodiversity tracking.

The decoupled architecture does not bottleneck downstream data consumption or high-volume data ingestion in Kafka. Producers (in this case, environmental sensor networks, satellite data feeds, or climate model outputs) and consumers (ML models for predictive analytics) operate independently. This means the data production rate doesn't affect the data processing rate, ensuring smooth operations even during traffic spikes, such as during extreme weather events or seasonal ecological changes. Decoupling enables scalability, which implies the ability to add new ML models, as well as flexibility because the existing workflow is left untouched, with the flexibility to include additional subjects-such as a new environmental data category-without slowing down any ongoing workflows. For example, one may use an emerging algorithm to predict carbon flux from forests by using the new consumers without disrupting ongoing processes for, say, tracking water quality.

Use of Kafka enables real-time monitoring and analytics in environmental informatics. The pipeline aids in the uninterrupted flow of data into Kafka topics whereby the machine learning models assimilate and analyze information in the real-time for prompt insight provision and supporting predictive analysis. This thus creates speedy decision-making making the whole system very suitable for examples like dynamic resource distribution in the case of precision agriculture or early alert mechanisms for natural catastrophes, real-time survey of wildlife migration trends, continuous monitoring of greenhouse gas emissions.

Compatibility with other Big Data technologies makes Kafka's utility in environmental monitoring systems much more valuable. With Apache Spark, it supports complex data processing and ML model training. With Apache Flink, it offers stream processing for continuous data related to the environment. Along with Hadoop HDFS, it supports long-term storage of historical environmental datasets, while with Elasticsearch, large volumes of environmental data can be indexed and queried.

Such an integrating capability builds excellent systems to the big environmental data platforms, suitable for real-time and batch processing requirements.

In the case of extended environmental monitoring, the data structures may change due to the addition of new sensors or advancements in measurement techniques. The schema evolution-related features of Kafka are very useful when used along with tools such as Apache Avro. It allows the system to adapt to the changes in data structures without disturbing already working consumers. This property is critical for an

extended environmental study, where backward compatibility should be preserved with already developed data. Within possibly sensitive or regulated environmental data, Kafka offers strong security functionalities, including SSL/TLS encryption for data-in-motion protection, SASL authentication mechanisms, and **Role-Based Access Control (RBAC) for fine-grained permissions**[20, 21]. Thus, these security measures assure that environmental data pipelines stay protected against data protection regulation contraventions while upholding integrity and confidentiality of the precious ecological data. In conclusion, it provides a more powerful, highly scalable, and flexible setup of infrastructure for tackling the complex challenge of modern eco-informatics that an application of a Kafka-based pipeline for Big Data in environmental monitoring and analysis would bring along. This opens the way to real-time data processing, safe data, smoother connectivity with advanced analytics tools, and better support for more precise and timely environmental decision making, enhancing effectiveness in conservation efforts as well as in climate change mitigations.

3.2 Machine Learning for Carbon Sequestration Prediction

This phase of this project focuses on the improvement of the accuracy of carbon sequestration predictions using machine learning algorithms[22]. Carbon sequestration, **the process of capturing atmospheric carbon dioxide and storing it in various natural sinks such as soil, forests, and oceans**, plays a crucial role in mitigating climate change. Traditional methods of estimating carbon sequestration are unreliable because they do not take into account differences in land management practices, soil types, and ecological factors among various ecosystems[23]. Conversely, the methods used for measuring carbon sequestration reliably are time consuming and often involve sending the soil samples to a lab.

To overcome these limitations, we developed ML models that we have trained on large datasets encompassing various environmental variables. We obtained the dataset from Kaggle, which is an open source platform that contains numerous datasets for various purposes. The dataset contains information such as soil type, bulk density, climate data (temperature, precipitation), vegetation cover, and land management practices (e.g., crop rotation, afforestation). These variables are important for the accurate prediction of the amount of carbon that can be sequestered in different ecosystems. Figure 3 illustrates the workflow of the machine learning model, highlighting the integration of these environmental variables for carbon sequestration prediction.

A comprehensive approach to preparing and engineering features for the SOC prediction model is what this study did. The process involved several key steps:

3.2.1 Feature Selection

Initially, we first chose a sub-set of relevant features from the source dataset [24, 25]. These features involved current as well as historical properties of the soil (pH, bulk density, clay), climatic conditions (temperature, precipitation) as well as management practices like fertilization, tillage type. All these have been chosen through domain knowledge and possible influences on SOC dynamics.

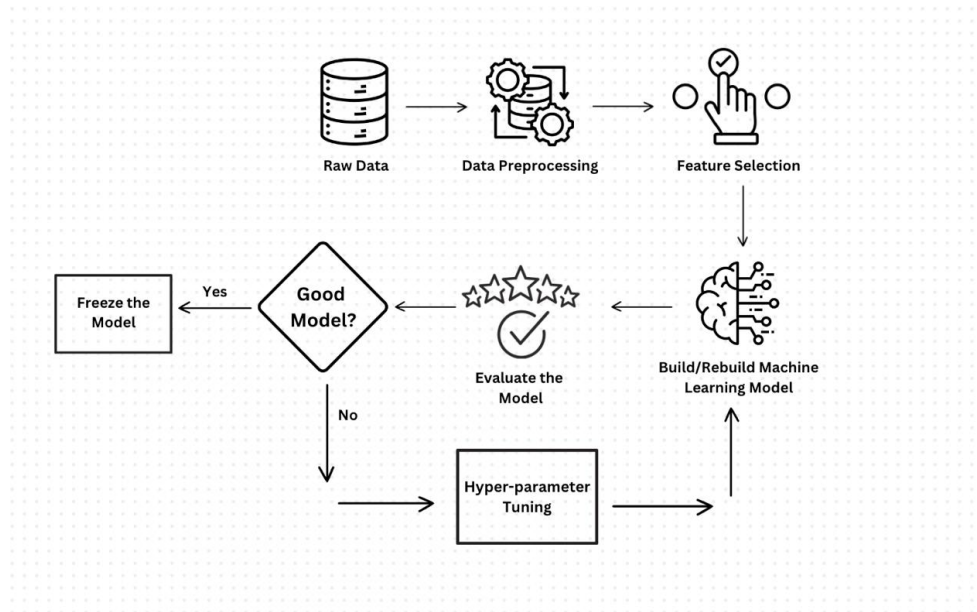


Fig. 3 Machine Learning Workflow

3.2.2 Correlation Analysis

To understand the relationships between variables and identify potential multi-collinearity, we conducted a correlation analysis[26]. We utilized Seaborn's heatmap visualization with a custom diverging color palette to illustrate the correlation matrix. Figure 4 presents the resulting correlation matrix, visualized using Seaborn's heatmap with a custom diverging color palette.

Through this analysis, we concluded that the 'SOC_Mg_ha', which is the carbon sequestration amount, was highly dependent on 8 features, namely current bulk density, previous bulk density recorded, current temperature, previously recorded temperature, current clay percentage, previously recorded clay percentage, current slit amount and previously recorded slit amount.

3.2.3 Data Cleaning and Missing Value Imputation

We addressed missing values using a combination of strategies[27]:

- For categorical variables (e.g., 'fertilizerYESNO', 'tillageYESNO', 'tillage_type'), we imputed missing values with the mode.
- For continuous variables, we employed different imputation methods based on the distribution of the data:
 - 1. Median imputation was used for 'SOC_Mg_ha_current', 'SOC_Mg_ha_previous', '%clay_previous', and '%sand_previous' to mitigate the effect of outliers.
 - 2. Mean imputation was applied to 'bulk.density_Mg_m3_current', 'Bulkdensity_previous', 'temperature_Celsius', '%clay_current', and '%silt_current'.

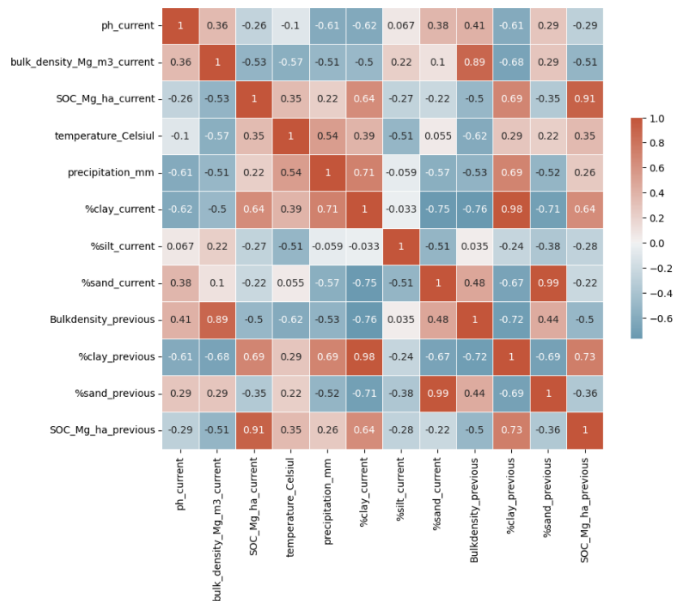


Fig. 4 Correlation Matrix

This approach ensured the preservation of the overall distribution of the variables while minimizing the impact of extreme values.

3.2.4 Outlier Removal

To enhance the robustness of our model, we implemented an outlier removal technique for the target variable 'SOC_Mg_ha_current'. We applied the Interquartile Range (IQR) method[28]:

- Calculated Q1 (25th percentile) and Q3 (75th percentile)
- Defined the IQR as $Q3 - Q1$
- Established lower and upper bounds as $Q1 - 1.5 \times IQR$ and $Q3 + 1.5 \times IQR$, respectively
- Removed data points falling outside these bounds

This step helped in reducing the influence of extreme values on our model's performance.

3.2.5 Feature Encoding

We encoded categorical variables using one-hot encoding before feeding into machine learning algorithms. We made use of 'fertilizerYESNO', 'tillageYESNO', and 'tillage.type'. After binary columns, we cleaned out the boolean values and replace them with numeric values so that it can be useful for our chosen algorithms as False: 0 and True: 1.

3.2.6 Train-Test Split

In summary, the data set preprocessed was divided into training and test subsets. Specifically, 80% of it was used in training while the remaining 20% was allocated to testing wherein a random state of 42 is used for the sake of reproducing the process. For model training and testing, 'SOC_Mg_ha_current' was copied from the feature set.

The detailed process of data preparation and feature engineering set a vital foundation for the development of a reliable model to predict soil organic carbon. Correcting missing data, removing anomalies, and proper encoding of categorical variables were all aimed at enhancing the model's ability to recognize subtle relationships between soil properties, environmental factors, and management practices in predicting soil organic carbon levels.

3.2.7 Model Selection and Training

Following the data preparation and feature engineering phase, the project proceeded with model selection and training. Several machine learning algorithms were tested and compared to determine the most effective model for predicting carbon sequestration. The algorithms considered included Random Forest (RF), XGBoost, and Support Vector Regressors (SVRs)[29–31] and Recursive Neural Network (RNN). These models were selected based on their demonstrated capability in other research papers in handling large and complex datasets, making them suitable candidates for this project.

The models were trained on the prepared historical data. To evaluate their predictive accuracy, performance metrics such as Root Mean Square Error (RMSE) and R-squared were employed. To ensure the models were not overfitting to the training data and could generalize well to unseen data, cross-validation techniques, specifically k-fold cross-validation, were utilized. This rigorous evaluation process aimed to identify the most robust and accurate model for carbon sequestration prediction.

3.2.8 Prediction and Optimization

After this training, the best-performing machine learning model, which turned out to be RNN, was selected in order to predict the ability of ecosystems to predict carbon sequestration. Such predictions were targeted to guide the land manager in better adaptation of their land-use policies to sequester carbon. In this regard, machine learning would provide an improved model suggesting optimum crop variety to be sown and proper means of tillage, increasing soil carbon capture by large percent.

This approach enabled the us to optimize the framework, which is combining machine learning model predictions with the constraints in land management. This approach serves as an optimization framework meant to deliver practical insights regarding how the carbon sequestration goal can be reconciled with other requirements on the use of the land to be economically feasible and beneficial environmentally.

3.2.9 Deployment

In this stage of the methodology, a predictive model was deployed in service form that could be used by landowners, environmental agencies, and others. This is so that users

could input their specific data regarding the environment and obtain actual predictions on the carbon sequestration capacity of their land.

It involved the development of an interface that was friendly in terms of multiple input formats but clear and interpretable to provide good outputs. It incorporated data validation checks to ensure that the quality and relevance of the provided user data were assured, and such ensured better reliability of the prediction.

It has been designed to be scalable so that it could accommodate thousands of concurrent queries for real-time forecasts. Such an approach is expected to arm the stakeholders with information on making informed decisions regarding the best land management practices that will ultimately enhance the potential for carbon sequestration at a broader scale.

By making this freely accessible and powerful tool, the project puts an effort to bridge the gap between sophisticated machine learning models and action-oriented decisions in land management toward possible acceleration of efforts to mitigate climate change through enhanced carbon sequestration.

3.3 Blockchain-Enabled Carbon Credit Trading

The subsequent stage of the project is developing a blockchain-based platform for trading in carbon credits. Carbon credits form part of a market-based approach that is employed in efforts aimed at reducing the emission of greenhouse gases. During this process, entities which remove or reduce carbon from the environment sell credits to other entities which are required to decrease their emissions. However, the current carbon credit market faces challenges in regard to transparency and verification that fuel skepticism over the credit's legitimacy and the true environmental benefits that these credits are able to provide.

This means that the proposed blockchain-based platform will essentially address these problems through a secure, decentralized, and transparent system for carbon credit trading. Blockchain technology, being immutable and secure, is the best option to overcome the carbon credit market challenges [32, 33]. We shall implement a permissioned blockchain, where only the participating parties such as landowners, buyers, and regulatory bodies will be allowed. In this manner, the transactions are traceable and verifiable but kept private and secure.

- The smart contracts [34, 35] autonomous agreements that have all the stipulations encoded in their code are the root element of our blockchain framework. These are going to provide the way for automating the entire verification process on carbon credits. Once carbon sequestration by any land owner is confirmed for an amount specified, the smart contract will produce the corresponding amount of carbon credit autonomously on the basis of the outputs of predictions from the ML model used. Credits can be issued only after fulfillment of specific conditions, such as verification from third parties of carbon sequestration data. All requirement for intermediaries is entirely dispensed with and fraud possibilities brought down to the bare minimum.

3.3.1 Blockchain-based Contract Transactions Algorithm

Algorithm Description

This is the algorithm to manage a list of transactions based on a blockchain contract transaction within a blockchain. Every transaction is represented by a TransferStruct that contains information, like sender, receiver, amount, and even more metadata.

The main function, addToBlockchain, creates and saves new transactions. It increases a global transaction counter, constructs a new TransferStruct object using the provided information and a current timestamp, adds it to the list of transactions, and fires a Transfer event to notify observers of the newly created transaction.

The helper functions which were introduced here are getAllTransactions and getTransactionCount. With these, one is allowed the query of the state without modifying it. It would help in recording the transactions and thus retrieve them correctly.

This algorithm will thus ensure that each transaction recorded by the algorithm is recorded accordingly and retrieved easily. Thus it is useful for any variety of blockchain applications which demands transparent and immutable history concerning transactions.

- Carbon Credits Tokenization : Every carbon credit will be tokenized, offering that precise digital asset, for safekeeping in the blockchain. Through this, tokens can afterwards be traded on the trading platform, thereby directly allowing a carbon credit buyer to buy a carbon credit from its seller. Tokenization is assured; each credit has full traceability history verifiable by checking instantly by any participant in that system.
- Trading Mechanism. Carbon credits will be sold and bought between buyers and sellers on the platform. Demand and supply will be fundamental factors in determining the price in any financial market, and the smart contract automates the transactions for this platform to provide a transparent trading mechanism. On closing of the trade, the blockchain will write the transaction for immutability and audibility.
- Security and Transparency: One of the key advantages of blockchain technology is its security. All transactions on the platform will be encrypted, and the decentralized nature of the blockchain ensures that no single entity can manipulate the data. Moreover, the transparent nature of the blockchain will allow stakeholders to verify the authenticity of carbon credits and track their origin.

Data Structures and Variables

- *transactionCounter*: integer, initially set to 0
- *transactions*: list of TransferStruct objects
- TransferStruct:
 - *sender*: address
 - *receiver*: address
 - *amount*: integer
 - *message*: string
 - *timestamp*: integer
 - *keyword*: string

Events

Define event:

$Transfer(sender, receiver, amount, message, timestamp, keyword)$ to be emitted upon successful transaction addition.

Main Algorithm

Algorithm 1 addToBlockchain

```

1: procedure ADDTOBLOCKCHAIN(receiver, amount, message, keyword)
2:   transactionCounter  $\leftarrow$  transactionCounter + 1
3:   newTransaction  $\leftarrow$  new TransferStruct
4:   newTransaction.sender  $\leftarrow$  getCurrentSender()
5:   newTransaction.receiver  $\leftarrow$  receiver
6:   newTransaction.amount  $\leftarrow$  amount
7:   newTransaction.message  $\leftarrow$  message
8:   newTransaction.timestamp  $\leftarrow$  getCurrentTimestamp()
9:   newTransaction.keyword  $\leftarrow$  keyword
10:  Append newTransaction to transactions
11:  Emit  $Transfer(newTransaction.sender, newTransaction.receiver,$ 
12:     $newTransaction.amount, newTransaction.message,$ 
13:     $newTransaction.timestamp, newTransaction.keyword)$ 
14: end procedure

```

Algorithm 2 getAllTransactions

```

1: procedure GETALLTRANSACTIONS return transactions
2: end procedure

```

Algorithm 3 getTransactionCount

```

1: procedure GETTRANSACTIONCOUNT return transactionCounter
2: end procedure

```

4 Results and Discussion

Fig. 5 shows that the RMSE values for predicting SOC_MG_HA which is our Soil Organic Carbon Content, vary across different models, with Deep Learning performing the worst, having RMSE of 18.64 and RNN achieving the best performance with RMSE of 15.32. Random Forest achieved an RMSE of 16.04 and SVR achieved an RMSE of 16.1. They exhibit similar results, while XGBoost has an RMSE of 15.34, which provides better accuracy than both. Overall, RNN and XGBoost deliver the most accurate predictions with the lowest RMSE, making them the most suitable models for this task, whereas Deep Learning shows the least reliable performance.

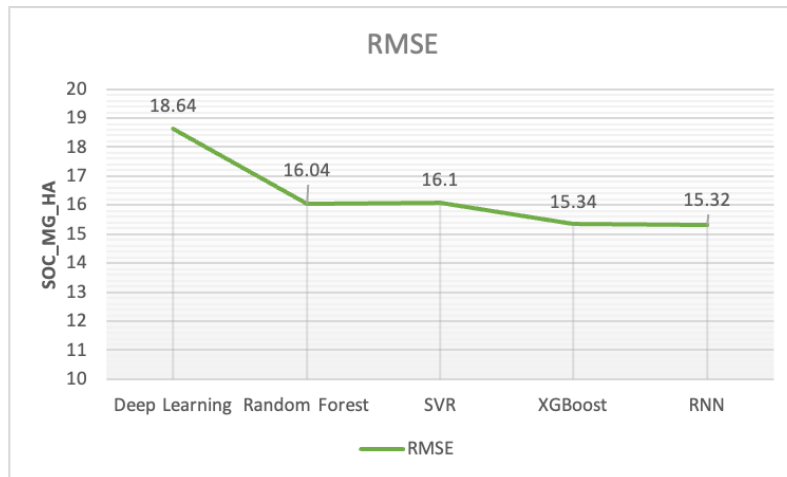


Fig. 5 RMSE

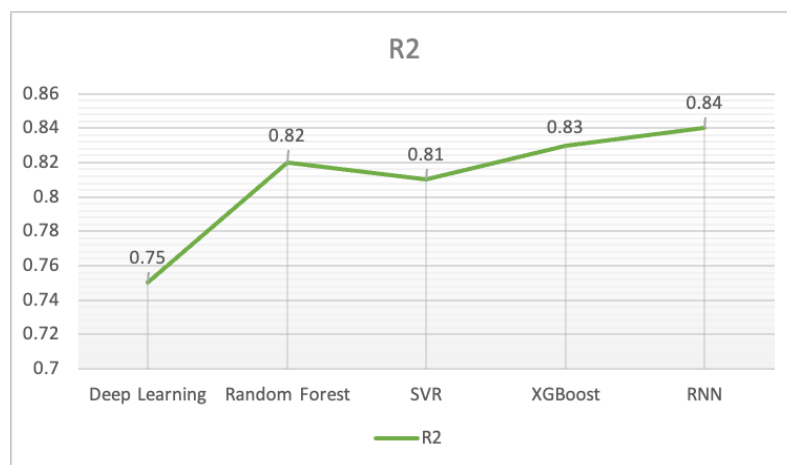


Fig. 6 Coefficient of Determination

Fig. 6 shows that the R^2 values, which indicate the goodness of fit for predicting SOC_MG_HA, vary across the models. Deep Learning has the lowest R^2 score of 0.75, suggesting a weaker fit to the data compared to other models. Random Forest shows improvement with an R^2 of 0.82, while SVR has a slightly lower score of 0.81. XGBoost performs better with an R^2 of 0.83, and RNN achieves the highest R^2 value of 0.84, indicating the best fit to the data. Thus, RNN and XGBoost show the strongest predictive abilities, while Deep Learning exhibits the weakest.

The platform integrates several key components to facilitate carbon credit trading:

- **Login Page:** The secure login system (Figure 7) allows users to access the platform by entering their credentials. Role-based access ensures personalized user experiences for farmers, industrialists, and other stakeholders.

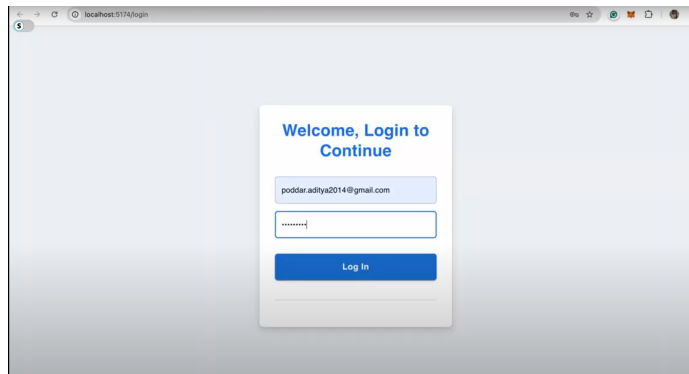


Fig. 7 Login Page

- **Soil Carbon Predictor:** Farmers input historical soil organic carbon (SOC) values into the ML-powered predictor (Figure 8). The model forecasts future SOC levels, determining how many carbon credits can be allocated based on sustainable farming practices.

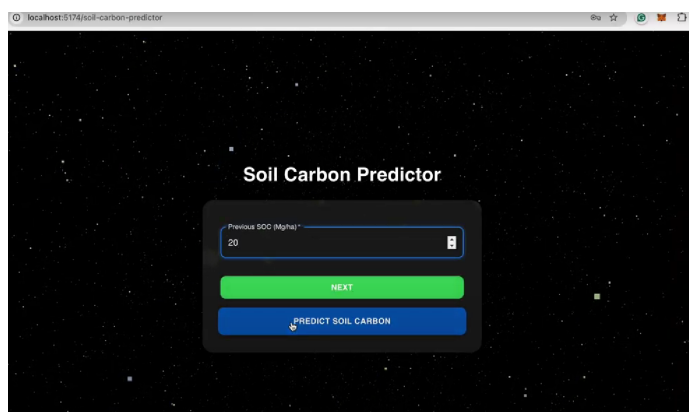


Fig. 8 User Interface for Prediction of Carbon Sequestration

- **Blockchain-Based Carbon Credit Exchange:** After the machine learning model predicts the soil organic carbon (SOC) levels, the platform calculates the corresponding number of carbon credits, which are then credited to the farmer's account. These credits are stored on the blockchain, ensuring that all transactions are immutable, transparent, and secure. The intermediaries also get eliminated due to this technology, which cuts down transaction costs and efficiency increases.

Farmers can trade their carbon credits with industrialists who want to offset their carbon emissions in the Carbon Credit X-Change (Figure 9). Smart contracts automatically execute these trades once pre-defined conditions are met, ensuring a trustless environment. This decentralized system guarantees that transactions are completed

without the need for direct human intervention, providing a reliable and tamper-proof mechanism for carbon trading.

Moreover, the blockchain ledger records each transaction as shown in Figure 10, making it auditable and fully traceable, thereby promoting accountability and trust among participants. This system promotes sustainable farming practices, as farmers are incentivized to adopt carbon sequestration methods, while industrialists benefit by meeting regulatory requirements and enhancing their environmental footprint.

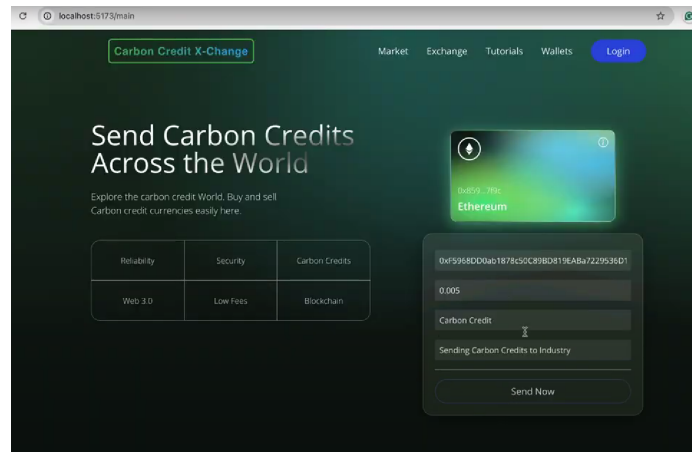


Fig. 9 User Interface for Carbon Credit Exchange

5 Conclusion

In a nutshell, the culmination of research has successfully developed the innovative framework for integrating technologies related to Big Data, Machine Learning, and Blockchains, focusing on massive problems such as carbon sequestration and carbon trading. The novelty in this line of research lies in an interdisciplinary methodology, which was founded by real-time data processing, predictive analytics, and secure digital trading systems. Meanwhile, our framework uses Kafka to make all the data streaming parallel for such machine learning models that already outperform traditional approaches being followed in carbon capture forecasting; therefore, more reliable and practical insights that a strategy of land management would gain more favorable carbon sequestration outcomes.

This is giving a clear, safe, and tamper-resistant trading framework for carbon credits. With the tokenization of carbon credits and the use of smart contracts, we have made the verification and trading processes easier, so also the fraud risk and the trust between participants are eliminated. Thus, it satisfies the environment as well as economics; therefore, it gives a scalable solution in accordance with the global sustainability goals.

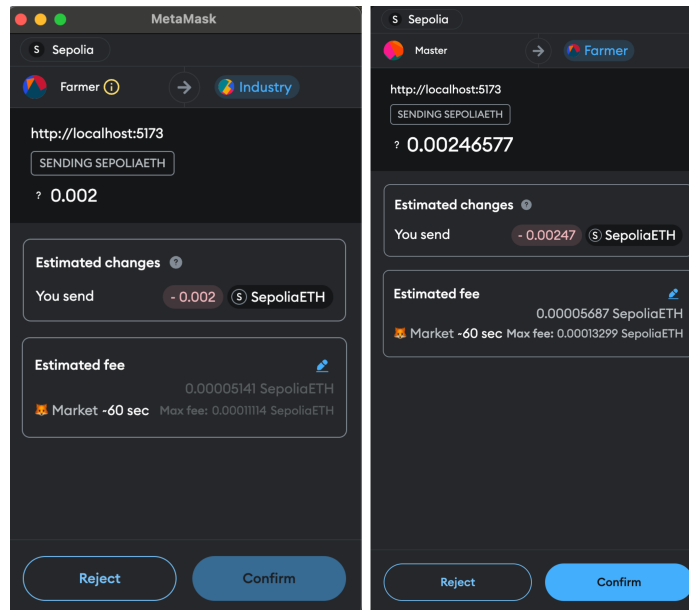


Fig. 10 Transactions between master-farmer and farmer-industry

6 Statements and Declarations

6.1 Competing Interests

Not Applicable

6.2 Funding

The authors declare that this research did not receive any specific grant from funding agencies in the public, commercial, or not-for-profit sectors. All work presented in this study was conducted independently by the authors, utilizing open-source platforms and test networks for research and development purposes.

6.3 Author contribution

All the authors contributed equally to this work.

6.4 Data Availability Statement

The dataset used for training the machine learning model is publicly available on the Kaggle website

6.5 Research Involving Human and /or Animals

Not Applicable

6.6 Informed Consent

Not Applicable

6.7 Conflict of Interest

The authors declare no conflicts of interest relevant to the content of this research.

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